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Severe fever with thrombocytopenia syndrome virus

Abstract

The present invention relates to a novel genotype of severe fever with thrombocytopenia syndrome viruses and use thereof as an immunogenic composition. The severe fever with thrombocytopenia syndrome viruses of the present invention are genetically different from conventional severe fever with thrombocytopenia syndrome viruses and are novel viruses taxonomically belonging to three sub-groups of genotype B. In view of the vaccine property that specific genotype viruses alone show only limited protective potential, the novel viruses of the present invention may be advantageously used as a vaccine having excellent cross-immunogenicity for SFTSV.

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USPC: None

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a Continuation Application of U.S. application Ser. No. 17/050,602 filed Oct. 26, 2020 which is a National Stage Application of International Application No. PCT/KR2019/004857 filed Apr. 23, 2019, claiming priority based on Korean Patent Application No. 10-2018-0047865 filed Apr. 25, 2018, the entire disclosures of which are incorporated herein by reference.

INCORPORATION BY REFERENCE OF SEQUENCE LISTING

(1) The content of the electronically submitted sequence listing, file name: Q288727_sequence listing as filed.XML; size: 76,505 bytes; and date of creation: Jul. 1, 2023, filed herewith, is incorporated herein by reference in its entirety.

TECHNICAL FIELD

(2) The present disclosure relates to a novel genotype of severe fever with thrombocytopenia syndrome virus and use thereof as an immunogenic composition.

BACKGROUND ART

(3) A severe fever with thrombocytopenia syndrome (SFTS) is accompanied by symptoms such as high fever, vomiting, diarrhea, thrombocytopenia, leukopenia and multiple organ failure, and is a serious disease with a mortality rate of 6% to 30% (Yu X J et al., N. Engl. J. Med. 2011; 364:1523-32; Ding F et al Clin Infect Dis 2013; 56: 1682-3).

(4) A causative pathogen of SFTS is SFTSV (severe fever thrombocytopenia syndrome virus), which belongs to Bunyaviridae family. Bunyaviridae family is a negative-strand RNA virus containing three segments. Bunyaviridae family includes five genera including *Orthobunyavirus*, *Hantavirus*, *Nairovirus*, *Phlebovirus* and *Tospovirus*. SFTSV belongs to the *Phlebovirus* genus which includes Rift valley fever virus. SFTSV was first reported in China in 2011 (Yu X J et al. *ibid*), and is a new variant virus that continues to outbreak not only in China, but also in Korea and Japan. SFTSV is a ball-shaped virus with a diameter of 80 nm to 100 nm. This virus carries three genes: a large (L) segment as a single-stranded negative sense RNA segment, a medium (M) segment, and a small (S) segment (NP, NS).

(5) The SFTS virus is known to spread via *Haemaphysalis longicornis* as a vector thereof, which spreads widely in Korea as well (Chae J S et al. J Vet Sci 2008; 9: 285-93; Kim C M et al. Appl Environ Microbiol 2006; 72: 5766-76). Seroconversion and viraemia of the SFTS virus have been found in domestic animals such as goats, sheep, cattle, pigs and dogs. It is believed that these animals act as intermediate vectors thereof in the area where SFTS virus spreads (Zhao L et al. Emerg Infect Dis 2013; 18: 963-5; Niu G et al. Emerg Infect Dis 2013; 19: 756-63). SFTSV is detected in the blood of patients, and the concentration of SFTSV is very high in blood of severely ill patients. Thus, human-to-human transmission thereof is possible via the blood (Tang X, Wu W, Wang H, et al. J Infect Dis 2013; 207: 736-739.).

(6) Antiviral agents for SFTSV have not been developed yet, and thus SFTS treatment is based on conservative therapy for organ failure such as blood transfusion and renal replacement therapy. In

China, ribavirin infusion has been introduced into the treatment guidelines since 2012. However, there was no difference in the mortality rate between the ribavirin-treated group and the non-administered group in the recently published treatment results. Therefore, a vaccine against SFTSV is required, but such a vaccine has not been developed yet.

DISCLOSURE

Technical Purpose

(7) A purpose of the present disclosure is to provide a novel genotype of severe fever with thrombocytopenia syndrome virus and an immunogenic composition containing the same.

Technical Solution

(8) To achieve the purpose, the present disclosure provides a novel severe fever with thrombocytopenia syndrome virus.

(9) Further, the present disclosure provides an immunogenic composition for prevention or treatment of the severe fever with thrombocytopenia syndrome.

(10) Further, the present disclosure provides antibodies against the severe fever with thrombocytopenia syndrome virus or an antigen thereof.

(11) Further, the present disclosure provides a diagnostic kit for the severe fever with thrombocytopenia syndrome virus.

(12) Further, the present disclosure provides a method to detect the severe fever with thrombocytopenia syndrome virus antibody.

(13) Further, the present disclosure provides a method for producing antiserum against the severe fever with thrombocytopenia syndrome virus.

(14) In addition, the present disclosure provides a method to provide information regarding diagnosis of the severe fever with thrombocytopenia syndrome.

Advantageous Effects

(15) The severe fever with thrombocytopenia syndrome virus according to the present disclosure is genetically different from the conventional thrombocytopenia virus and is a novel virus as systematically subdivided from a B genotype. Thus, the novel virus according to the present disclosure may be usefully used as vaccines with excellent cross immunogenicity against SFTSV, due to characteristics of a vaccine that only a specific genotype virus exhibits limited protective ability.

Description

DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a phylogenetic tree showing three novel viruses according to the present disclosure and SFTSV L gene isolated from China, Japan, and Korea.

(2) FIG. 2 is a phylogenetic tree showing the three novel viruses according to the present disclosure and SFTSV M gene isolated from China, Japan, and Korea.

(3) FIG. 3 is a phylogenetic tree showing the three novel viruses according to the present disclosure and SFTSV S (NP) gene isolated from China, Japan, and Korea.

(4) FIG. 4 is a phylogenetic tree showing the three novel viruses according to the present disclosure and SFTSV S (NS) gene isolated from China, Japan, and Korea.

(5) FIG. 5 shows amino acid mutation sites in L and M genes of three novel viruses of genotypes B-1 (CB3/2016), B-2 (CB7/2017) and B-3 (CB6/2016) according to the present disclosure.

(6) FIG. 6 is a diagram that identifies vaccine effects of three novel viruses of genotypes B-1 (CB3/2016), B-2 (CB7/2017) and B-3 (CB6/2016) according to the present disclosure.

MODES OF THE INVENTION

(7) Hereinafter, the present disclosure will be described in detail based on implementations according to the present disclosure with reference to the accompanying drawings. However, the

following implementations are presented as only examples of the present disclosure. When it is determined that detailed description of a well-known component or configuration may unnecessarily obscure the gist of the present disclosure, the detailed description may be omitted. The implementations do not limit the present disclosure. The present disclosure may be variously modified and applied within the scope interpreted based on the claims to be described later.

(8) Further, terms (terminologies) used in this specification are used to properly describe preferred Example of the present disclosure, and may vary according to a user's or operator's intention or a practice of the field to which the present disclosure belongs. Accordingly, definitions of these terms should be made based on contents throughout the present specification. It will be further understood that the terms “comprises”, “comprising”, “includes”, and “including”, “containing” and “contains”, etc. when used in this specification, specify the presence of the stated elements, and/or components, but do not preclude the presence or addition of one or more other elements, components, and/or portions thereof.

(9) In one aspect, the present disclosure relates to the severe fever with thrombocytopenia syndrome virus (SFTSV) in which a 1447-th amino acid of a protein expressed in ORF (6255 bp) of an L gene thereof is valine or a 1913-rd amino acid thereof is lysine, wherein a 83-rd amino acid of a protein expressed in ORF (3222 bp) of an M gene thereof is tyrosine, or a 404-th amino acid thereof is threonine or a 904-th amino acid thereof is valine. In one Example of the present disclosure, this virus was named B-1 (CB3).

(10) In one implementation, the severe fever with thrombocytopenia syndrome virus belongs to genotype B-1 and may be a virus in which isoleucine as a 1447-th amino acid of a protein expressed in ORF (6255 bp) of an L gene of a virus conventionally classified as a genotype B is substituted with valine, and arginine as a 1913-rd amino acid thereof is substituted (SEQ ID NO: 13) with lysine, and isoleucine as a 904-th amino acid of an M gene thereof is substituted (SEQ ID NO: 14) with valine.

(11) In one implementation, the severe fever with thrombocytopenia syndrome virus may contain an L gene containing a base sequence represented by SEQ ID NO: 1, an M gene containing a base sequence represented by SEQ ID NO: 2, and an S gene containing NP containing a base sequence represented by SEQ ID NO: 3 and NS containing a base sequence represented by SEQ ID NO: 4.

(12) In one implementation, the severe fever with thrombocytopenia syndrome virus may contain an amino acid sequence represented by SEQ ID NO: 13 expressed in ORF (6255 bp) of an L gene, an amino acid sequence represented by SEQ ID NO: 14 expressed in ORF (3222 bp) of an M gene, and an amino acid sequence represented by SEQ ID NO: 15 expressed in ORF of an NP gene, and an amino acid sequence represented by SEQ ID NO: 16 expressed in ORF of an NS gene.

(13) In one aspect, the present disclosure relates to a severe fever with thrombocytopenia syndrome virus in which a 1447-th amino acid of a protein expressed in ORF (6255 bp) of an L gene thereof is isoleucine or a 1913-rd amino acid thereof is arginine, wherein a 83-rd amino acid of a protein expressed in ORF (3222 bp) of an M gene is phenylalanine, a 404-th amino acid thereof is threonine or a 904-th amino acid thereof is isoleucine. In one Example of the present disclosure, the virus was named B-2 (CB4).

(14) In one implementation, the severe fever with thrombocytopenia syndrome virus belongs to genotype B-2, and may be a virus in which tyrosine as an 83-rd amino acid of an M gene of a virus conventionally classified as a genotype B has been replaced with phenylalanine.

(15) In one implementation, the severe fever with thrombocytopenia syndrome virus may contain an L gene containing a base sequence represented by SEQ ID NO: 5, an M gene containing a base sequence represented by SEQ ID NO: 6, and an S gene containing a NP containing a base sequence represented by SEQ ID NO: 7 and a NS containing a base sequence represented by SEQ ID NO: 8.

(16) In one implementation, the severe fever with thrombocytopenia syndrome virus may contain an amino acid sequence represented by SEQ ID NO: 17 expressed in ORF of an L gene, an amino acid sequence represented by SEQ ID NO: 18 expressed in ORF of an M gene, an amino acid

sequence represented by SEQ ID NO: 19 expressed in ORF of an NP gene and an amino acid sequence represented by SEQ ID NO: 20 expressed in ORF of an NS gene.

(17) In one aspect, the present disclosure relates to a severe fever with thrombocytopenia syndrome virus in which a 1447-th amino acid of a protein expressed in ORF (6255 bp) of an L gene thereof is isoleucine or a 1913-rd amino acid thereof is arginine, wherein a 83-rd amino acid of a protein expressed in ORF (3222 bp) of an M gene thereof is tyrosine, a 404-th amino acid thereof is alanine or a 904-th amino acid thereof is isoleucine. In one Example of the present disclosure, the virus was named B-3 (CB1).

(18) In one implementation, the severe fever with thrombocytopenia syndrome virus belongs to genotype B-3, and may be a virus in which threonine as a 404-th amino acid of an M gene of a virus conventionally classified as a genotype B is substituted with alanine.

(19) In one implementation, the severe fever with thrombocytopenia syndrome virus may contain an L gene containing a base sequence represented by SEQ ID NO: 9, an M gene containing a base sequence represented by SEQ ID NO: 10, and an S gene containing NP containing a base sequence represented by SEQ ID NO: 11 and NS containing a base sequence represented by SEQ ID NO: 12.

(20) In one implementation, the severe fever with thrombocytopenia syndrome virus may contain an amino acid sequence represented by SEQ ID NO: 21 expressed in ORF of an L gene, an amino acid sequence represented by SEQ ID NO: 22 expressed in ORF of an M gene, an amino acid sequence represented by SEQ ID NO: 23 expressed in ORF of an NP gene, and an amino acid sequence represented by SEQ ID NO: 24 expressed in ORF of an NS gene.

(21) In one example of the present disclosure, it was revealed based on a result of genetic analysis of the severe fever with thrombocytopenia syndrome virus as isolated that a gene sequence thereof was different from that of the severe fever with thrombocytopenia syndrome virus as previously known. It was revealed based on a result of systematically classifying the severe fever with thrombocytopenia syndrome virus according to the present disclosure that a genotype thereof is subdivided into at least three genotypes other than a single B genotype group as previously known.

(22) The term “substitution” as used in the present disclosure refers to replacement of one or more amino acids or nucleotides by other amino acids or nucleotides, respectively.

(23) The severe fever with thrombocytopenia syndrome virus according to the present disclosure is a negative single-stranded RNA virus and belongs to Bunyaviridae family and to *phlebovirus* genus. The severe fever with thrombocytopenia syndrome virus according to the present disclosure is a spherical virus with a diameter of 80 nm to 100 nm and spreads via *Haemaphysalis longicornis* as a vector thereof. A genome thereof includes a large (L) segment, a medium (M) segment, and a small (S) segment and codes six proteins including RNA dependent RNA polymerase (RdRp), glycoprotein precursor (M), glycoprotein N (Gn), glycoprotein C (Gc), nucleocapsid protein (NP), and non-structural protein (NS). In a negative or antisense strand (sense encoding a viral protein or antisense against a positive strand), a protein or gene is encoded as antisense. For expression of a gene into a protein, a sense or positive strand RNA is generated, and then translation therefrom is performed such that the protein is produced.

(24) In one aspect, the present disclosure relates to an immunogenic composition for prevention or treatment of a severe fever with thrombocytopenia syndrome, the composition containing the severe fever with thrombocytopenia syndrome virus or an antigen thereof as an active ingredient.

(25) In one implementation, the immunogenic composition according to the present disclosure may contain an inactivated severe fever with thrombocytopenia syndrome virus and a pharmaceutically acceptable carrier or adjuvant.

(26) In one implementation, the immunogenic composition may be a vaccine composition, wherein a form thereof may be selected from the group consisting of live vaccine, killed vaccine, subunit vaccine produced using a gene of an attenuated severe fever with thrombocytopenia syndrome virus, vector vaccine, chimeric vaccine, DNA vaccine, and RNA vaccine.

(27) In one implementation, the immunogenic composition may contain, as an active ingredient, a

severe fever with thrombocytopenia syndrome virus containing an L gene containing a base sequence represented by SEQ ID NO: 1, an M gene containing a base sequence represented by SEQ ID NO: 2, and an S gene containing a base sequence represented by SEQ ID NO: 3; a severe fever with thrombocytopenia syndrome virus containing an L gene containing a base sequence represented by SEQ ID NO: 4, an M gene containing a base sequence represented by SEQ ID NO: 5, and an S gene containing a base sequence represented by SEQ ID NO: 6; and a severe fever with thrombocytopenia syndrome virus containing an L gene containing a base sequence represented by SEQ ID NO: 7, an M gene containing a base sequence represented by SEQ ID NO: 8, and an S gene containing a base sequence represented by SEQ ID NO: 9; or antigens thereof.

(28) For preparation of the immunogenic composition (i.e., a vaccine) according to the present disclosure, the virus or an antigen thereof according to the present disclosure is transformed into a physiologically acceptable form. This may be done based on experiences of preparing a vaccine used for vaccination against influenza (disclosed by Stickl, H. et al. [1974] Dtsch. med. Wschr. 99, 2386-2392). For the preparation of vaccine injections, for example, virus particles are lyophilized in 100 ml of phosphate-buffered saline (PBS) under the presence of 1% human albumin and 2% peptone in ampoules, preferably in glass ampoules. Alternatively, vaccine injections may be produced by sequential freeze-drying of the virus in the formulation. This formulation may contain additional additives such as mannitol, dextran, sugar, glycine, lactose or polyvinylpyrrolidone or antioxidants or inert gases, stabilizers or other adjuvants such as recombinant proteins suitable for in vivo administration (e.g. human serum albumin). The glass ampoule may then be sealed and stored at a temperature between 4° C. and room temperature for several months. However, unless otherwise required, the ampoules may preferably be stored below -20° C.

(29) For vaccination or treatment, the lyophilisate may be dissolved in 0.1 ml to 0.5 ml of an aqueous solution, preferably physiological saline or tris buffer, and then may be administered to a subject systemically or locally, i.e. in parenteral, subcutaneous, intramuscular manner or via other routes of administration known to those skilled in the art. A dosage form, dosage and frequency of administration thereof may be optimized by a person skilled in the art in a known manner. However, most commonly, patients receive a second vaccination about a month to 6 weeks after a first vaccination.

(30) In the present disclosure, the term “prevention” refers to any action that inhibits or delays the occurrence, spread and recurrence of the severe fever with thrombocytopenia syndrome by administration of the immunogenic composition according to the present disclosure.

(31) The term “treatment” as used in the present disclosure refers to any action that reduces or beneficially alters the symptoms of the severe fever with thrombocytopenia syndrome and complications thereof via the administration of the immunogenic composition according to the present disclosure. A person with ordinary knowledge in the technical field to which the present disclosure belongs refers to the data presented by the Korean Medical Association, etc. to know the exact criteria about the disease to which the composition according to the present disclosure is effective, and to determine degrees of the improvement and treatment.

(32) The term “therapeutically effective amount” used in combination with an active ingredient in the present disclosure refers to an amount effective for preventing or treating the severe fever with thrombocytopenia syndrome. The therapeutically effective amount of the composition according to the present disclosure may vary depending on several factors, such as administration method, target site, and patient's condition. Therefore, when the composition is used for the human body, the dosage should be determined as an appropriate amount in consideration of safety and efficiency. The skilled person may estimate the amount to be used for humans from an effective amount determined through animal experiments. Factors to consider when determining the effective amount are described, for example, in Hardman and Limbird, eds., Goodman and Gilman's The Pharmacological Basis of Therapeutics, 10th ed. (2001), Pergamon Press; And E. W. Martin ed., Remington's Pharmaceutical Sciences, 18th ed. (1990), Mack Publishing Co.

(33) A pharmaceutical composition according to the present disclosure is administered in a pharmaceutically effective amount. As used in the present disclosure, the term “pharmaceutically effective amount” refers to an amount sufficient to treat the severe fever with thrombocytopenia syndrome at a reasonable benefit/risk ratio applicable to medical treatment and not to cause side effects. The effective dose level may be determined based on factors including the patient's health status, type of transplantation, severity, activity of the drug, sensitivity to the drug, method of administration, time of administration, route of administration and rate of excretion, duration of treatment, drugs used in combination or simultaneously, and other factors well known in the medical field. The composition according to the present disclosure may be administered as an individual therapeutic agent or administered in combination with other therapeutic agents or may be administered sequentially or simultaneously with a conventional treatment agent, or may be administered single or multiple times. Considering all of the above factors, it is important to administer the amount by which the maximum effect may be obtained at the minimum amount without side effects. This amount may be easily determined by a person skilled in the art.

(34) The pharmaceutical composition according to the present disclosure may contain carriers, diluents, excipients or a combination of two or more thereof commonly used in biological preparations. As used in the present disclosure, the term “pharmaceutically acceptable” refers to characteristics that a composition is not toxic to cells or humans as exposed to the composition. The carrier is not particularly limited as long as the carrier is suitable for delivery of the composition to a target site in vivo. The carrier may include, for example, compounds described in Merck Index, 13th ed., Merck & Co. Inc., saline, sterile water, Ringer's solution, buffered saline, dextrose solution, maltodextrin solution, glycerol, ethanol, and mixtures of one or more of these components. If necessary, other conventional additives such as antioxidants, buffers, and bacteriostatic agents may be added thereto. Further, when additionally adding a diluent, a dispersant, a surfactant, a binder, and a lubricant to the composition, the composition may be formulated into a formulation for injection such as an aqueous solution, a suspension, an emulsion, a pill, a capsule, a granule or a tablet. Furthermore, the composition may be preferably formulated based on each disease or component using a method appropriate in the art or by a method disclosed in Remington's Pharmaceutical Science (Mack Publishing Company, Easton PA, 18th, 1990).

(35) In one implementation, the pharmaceutical composition may be formulated into at least one selected from the group including oral dosage forms, external preparations, suppositories, sterile injectable solutions and sprays. Oral or injection formulations are more preferred.

(36) The term “administration” as used in the present disclosure means providing a predetermined substance to a subject or patient in any appropriate way. Depending on the intended method, parenteral administration (for example, an injection formulation being applied in intravenous, subcutaneous, intraperitoneal manner or topically) or oral administration may be possible. The dosage range varies depending on the patient's weight, age, sex, health status, diet, administration time, administration method, excretion rate, and severity of disease. Liquid formulations for oral administration of the composition according to the present disclosure include suspensions, liquid solutions, emulsions, syrups, etc. In addition to water and liquid paraffin, which are commonly used simple diluents, various excipients such as wetting agents, sweeteners, fragrances, and preservative may be contained therein together. Formulations for parenteral administration include sterile aqueous solutions may include non-aqueous solvents, suspensions, emulsions, lyophilized formulations, suppositories, and the like. The pharmaceutical composition according to the present disclosure may be administered using any device capable of delivering the active substance to the target cell. Preferred modes of administration and formulations may be intravenous injections, subcutaneous injections, intradermal injections, intramuscular injections, drop injections and the like. Injectables may be prepared using aqueous solvents such as physiological saline and Ringer solutions, or non-aqueous solvents such as vegetable oils, higher fatty acid esters (e.g., oleic acid ethyl, etc.), alcohols (e.g., ethanol, benzyl alcohol, propylene glycol, glycerin, etc.). It may contain

a pharmaceutical carrier such as stabilizers to prevent deterioration (e.g. ascorbic acid, sodium hydrogen sulfite, sodium pyro sulfite, BHA, tocopherol, EDTA, etc.), an emulsifier, a buffer for pH control, and a preservative for preventing the growth of microorganisms (e.g., phenyl mercury nitrate, thimerosal, benzalkonium chloride, phenol, cresol, benzyl alcohol, etc.).

(37) As used in the present disclosure, the term “subject” refers to monkeys, cows, horses, sheep, pigs, chickens, turkeys, quails, cats, dogs, mice, rats, rabbits or guinea pigs, or humans who have the severe fever with thrombocytopenia syndrome. The “specimen” may be whole blood, plasma, serum, urine or saliva isolated therefrom.

(38) The pharmaceutical composition according to the present disclosure may further contain pharmaceutically acceptable additives. The pharmaceutically acceptable additives include starch, gelatinized starch, microcrystalline cellulose, lactose, povidone, colloidal silicon dioxide, calcium hydrogen phosphate, lactose, mannitol, malt, arabic rubber, pregelatinized starch, corn starch, powdered cellulose, hydroxypropyl cellulose, opadry, sodium starch glycolate, lead carnauba, synthetic aluminum silicate, stearic acid, magnesium stearate, aluminum stearate, calcium stearate, sucrose, dextrose, sorbitol, talc, and the like. The pharmaceutically acceptable additive according to the present disclosure is preferably contained in 0.1 parts by weight to 90 parts by weight based on the composition, but is not limited thereto.

(39) In one aspect, the present disclosure relates to an antibody produced in response to immunization using the virus or antigen thereof according to the present disclosure.

(40) The antibody is not only in the form of a whole antibody, but also includes a functional fragment of an antibody molecule. The whole antibody has a structure having two full-length light chains and two full-length heavy chains, and each light chain is connected to a heavy chain via a disulfide bond. A functional fragment of an antibody molecule refers to a fragment that has an antigen-binding function. Examples of antibody fragments may include (i) a Fab fragment composed of a variable region (VL) of a light chain and a variable region (VH) of a heavy chain, a constant region (CL) of a light chain, and a first constant region (CH1) of a heavy chain; (ii) an Fd fragment composed of VH and CH1 domains; (iii) an Fv fragment composed of VL and VH domains of a single antibody; (iv) a dAb fragment composed of a VH domain (Ward E S et al., Nature 341:544-546 (1989)); (v) an isolated CDR region; (vi) an F(ab')₂ fragment as a bivalent fragment containing two linked Fab fragments; (vii) a single chain Fv molecule (scFv) bound via a peptide linker that binds the VH domain and the VL domain to each other to form an antigen binding site; (viii) a bispecific single-chain Fv dimer (PCT/US92/09965); and (ix) a diabody as a polyvalent or multispecific fragment produced by gene fusion (WO94/13804), and the like.

(41) In one aspect, the present disclosure relates to a diagnostic kit for the severe fever with thrombocytopenia syndrome virus, the kit containing the severe fever with thrombocytopenia syndrome virus or an antigen thereof, or an antibody against the same.

(42) In one implementation, the kit may contain a virus sample containing the virus according to the present disclosure and a reagent for detecting an antigen-antibody complex. The reagent for detecting the antigen-antibody complex includes reagents for radioimmunoassay, ELISA (Enzyme linked immunosorbent assay) or immunofluorescence analysis.

(43) In one embodiment, the detection of the antigen-antibody complex may be achieved using an Ouchterlony plate simply detecting the antibody and/or antigen via the antigen-antibody binding, western blot, and immuno electrophoresis such as Crossed IE, Rocket IE, Fused Rocket IE, and Affinity IE. Reagents or substances as used in this method are known. This may be detected, for example, via antigen-antibody reactions, or substrates, nucleic acids or peptide aptamers that specifically bind to antigens, or reactions with receptors, ligands, or cofactors interacting with the complex or using mass spectrometry. The reagent or substance that specifically interacts or binds to the antigen-antibody complex of the present application may be used in a chip method or in combination with nanoparticles. The immunoassay or immunostaining method is described in Enzyme Immunoassay, E. T. Maggio, ed., CRC Press, Boca Raton, Florida, 1980; Gastra, W.,

Enzyme-linked immunosorbent assay (ELISA), in *Methods in Molecular Biology*, Vol. 1, Walker, J. M. ed., Humana Press, NJ, 1984. Analyzing the intensity of the final signal by the above-described immunoassay process, that is, performing signal contrast with a normal sample may diagnose absence or presence of the infection of the disease.

(44) In one aspect, the present disclosure relates to a diagnostic composition containing the severe fever with thrombocytopenia syndrome virus or antigen thereof, or an antibody against the antigen.

(45) The compounds according to the present disclosure as used in the diagnostic composition are preferably labeled detectably. Various methods available for labeling biomolecules are well known to those skilled in the art and are considered within the category according to the present disclosure. The methods are described in Tijssen, 'Practice and theory of enzyme immuno assays', Burden, R H and von Knippenburg (Eds), Volume 15 (1985), 'Basic methods in molecular biology'; Davis L G, Dibmer M D; Battey Elsevier (1990), Mayer et al., (Eds) 'Immunochemical methods in cell and molecular biology' Academic Press, London (1987), or in the series 'Methods in Enzymology', Academic Press, Inc.

(46) There are many other marking methods and markers known to the skilled person. Examples of the types of markers that may be used in the present disclosure may be enzymes, radioactive isotopes, colloidal metals, fluorescent compounds, chemiluminescent compounds, and bioluminescent compounds.

(47) Commonly used markers include fluorescent substances (e.g., fluorescein, rhodamine, Texas red, etc.), enzymes (e.g. horseradish peroxidase, β -galactosidase, alkaline phosphatase), radioactive isotopes (e.g., ^{32}P or ^{125}I), biotin, digoxigenin, colloidal metal, chemiluminescent or bioluminescent compounds (e.g., dioxetane, luminol or acridinium). The marking methods such as methods based on covalent bonding of enzymes or biotinyl groups, iodination, phosphorylation, and biotinylation are well known in the art.

(48) The detection methods may include autoradiography, fluorescence microscopy, direct and indirect enzyme reactions, etc. but is not limited thereto. A commonly used detection assay may be a radioactive isotope or non-radioactive isotope method. These may include Western blotting, overlay-analysis, Radioimmuno Assay (RIA) and Immuno Radioimmunometric Assay (IRMA), Enzyme Immuno Assay (EIA), Enzyme Linked Immuno Sorbent Assay (ELISA), Fluorescent Immuno Assay (FIA), and Chemiluminescent Immune Assay (CLIA).

(49) In one aspect, the present disclosure relates to a method for detecting a severe fever with thrombocytopenia syndrome virus antibody, the method including contacting a sample isolated from a specimen with a virus or antigen thereof according to the present disclosure under a condition in which an antigen/antibody complex is able to be formed; and detecting formation of an antigen/antibody complex.

(50) In one aspect, the present disclosure relates to a method for producing antiserum against the severe fever with thrombocytopenia syndrome virus in a non-human animal, the method including administering the virus or antigen thereof according to the present disclosure to the non-human animal at an amount effective to induce an immune response; and collecting antiserum or plasma containing an antibody against the severe fever with thrombocytopenia syndrome virus.

(51) In one aspect, the present disclosure relates to a method for providing information regarding the diagnosis of the severe fever with thrombocytopenia syndrome, the method including contacting a sample isolated from a specimen with the virus or antigen thereof according to the present disclosure to form an antigen-antibody complex; and detecting the formation of the complex.

EXAMPLES

(52) The present disclosure is described in more detail based on following Examples. However, the following Examples are intended only for specifying the present disclosure, and the present disclosure is not limited thereto.

Example 1

Virus Isolation

(53) Blood from patients who visited university hospitals and were suspected of having symptoms of severe fever thrombocytopenia syndrome, and bloods from animals (goat and abandoned dogs) suspected of having symptoms of severe fever thrombocytopenia syndrome, and wild mite homogenate were used to identify whether the SFTSV (severe fever thrombocytopenia syndrome virus) thereof is positive/negative via real-time PCR, PCR and ELISA analysis. Specifically, a day before virus infection, VeroE6 cells were dispensed into a 12-well plate, cultured so that the cell density exceeded 60%, and the cells were washed with PBS. The cells were treated with 300 µl of serum from a suspected infected patient (serum obtained by centrifuging whole blood at 3000 rpm for 20 minutes) for 1 hour to infect the cells. After the infection, the serum was removed, and the cells were washed with PBS, and then the cells were exchanged with 1% FBS DMEM medium and cultured in 1% FBS DMEM medium for 2 weeks. The two weeks later, RT-PCR (identification via real-time PCR after reverse transcription) and immune fluorescence assay (in which a mouse SFTSV NP antibody produced in a laboratory was used as the primary antibody and the antibody conjugated with FITC was used as the secondary antibody) were used to identify the presence or absence of the virus isolation. When the virus was not isolated, the virus was isolated by infecting another VeroE6 cell with the first infected supernatant. The isolated viruses were named as CB3/2016, CB7/2017 and CB6/2016.

Example 2

Genetic Analysis of Isolated Virus

(54) The viruses CB3/2016, CB7/2017 and CB6/2016 isolated using Vero E6 cells were respectively reverse-transcribed, and then subjected to PCR, and NGS (next generation sequencing) to identify L, M, S (NP, NS) whole gene sequences thereof. Specifically, RNA was extracted from each virus, and cDNA was produced through reverse-transcription PCR. Subsequently, the L, M, and S genes of each SFTS virus were subjected to PCR to obtain each whole gene. The NGS method was used for gene sequence analysis. The L, M, and S genes of each virus were subjected to tagmentation and index PCR using an illumina nextera XT kit according to the protocol provided from illumina. Afterwards, Fasta Q file of the final sample was generated using the illumina miniseq equipment. The whole gene sequence of the generated file was analyzed using the CLC main workbench program. We integrated the identified gene sequences with the genes of conventional viruses isolated in Korea, China and Japan, and then performed genetic analysis. Thus, it was identified that the SFTSVs CB1, CB3 and CB4 as isolated according to the present disclosure are new genotypes of SFTSV genetically different from the genes of the viruses currently isolated in China or Korea (Yu X J et al., N. Engl. J. Med. 2011) and the virus first isolated in Korea (Gangwon/2012). In addition, it was identified based on a result of a phylogenetic gene analysis of L, M, and S (NP, NS) genes of viruses according to the present disclosure and viruses isolated from Korea, China, and Japan using the MEGA 7.0 program, that the viruses according to the present disclosure were close to the group B which most of the domestic isolated viruses belong to, but the genes L, M and S (NP and NS) thereof are different from those of the conventional viruses. Thus, we identified that the viruses according to the present disclosure is subdivided into at least 3 or more groups (FIG. 1 to FIG. 4). Accordingly, the three genotype SFTSVs isolated according to the present disclosure as subdivided were named B-1 (CB3/2016), B-2 (CB7/2017) and B-3 (CB6/2016), respectively.

Example 3

Identification of Differences in Amino Acid Sequence of L, M and S Genes by Viruses

(55) 3-1. L Gene Amino Acid Sequence Difference

(56) It was identified based on a result of analyzing an amino acid sequence based on the open reading frame (ORF) of each of the L, M and S genes of the viruses of the novel subdivided genotypes B-1, B-2 and B-3 isolated according to the present disclosure, that the L and M genes were different from those in the conventional virus. Specifically, ORF (6255 bp) of an L gene in the

B genotype encodes RdRp. In the genotype B-1 virus according to the present disclosure, the 1447-th amino acid of RdRp was valine or the 1913-rd amino acid thereof was lysine. In the genotype B-2 virus and B-3 virus according to the present disclosure, the 1447-th amino acid of RdRp was isoleucine or the 1913-rd amino acid thereof was arginine (FIG. 5A and Table 1).

(57) TABLE-US-00001 TABLE 1 Amino L gene M gene Acids site 1447 1913 83 404 904 B-1 type Val Lys Tyr Thr Val B-2 type Ile Arg Phe Thr Ile B-3 type Ile Arg Tyr Ala Ile

3-2. M Gene Amino Acid Sequence Difference

(58) It was identified based on a result of analyzing an amino acid sequences of the L, M and S genes of the viruses of the new subdivided genotypes B-1, B-2 and B-3 isolated in accordance with the present disclosure, the L and M genes were different from those in the conventional virus. Specifically, ORF (3222 bp) of an M gene in the B genotype encodes Gn and Gc proteins. According to the present disclosure, in the genotype B-1 virus, the 83-rd amino acid of ORF of an M gene was tyrosine or the 404-th amino acid thereof was threonine or the 904-th amino acid thereof was valine. In the genotype B-2 virus, the 83-rd amino acid of ORF of an M gene was phenylalanine, the 404-th amino acid thereof was threonine or the 904-th amino acid thereof was isoleucine. In the genotype B-3 virus, the 83-rd amino acid of ORF of an M gene was tyrosine, the 404-th amino acid thereof was alanine, or the 904-th amino acid thereof was isoleucine (FIG. 5B and Table 1).

Example 4

Comparison of Gene Homology Based on Genotype

(59) SFTSVs currently isolated in Korea and SFTSVs of the subdivided genotypes B-1, B-2 and B-3 according to the present disclosure were subjected to genetic analysis. The homology of L, M and S genes based on each genotype was compared and analyzed. As a result, about 96% to 100% of the gene homology (nucleotide level) between the viruses belonging to the same genotype was observed, while 91% to 97% of a relatively low gene homology between different genotype viruses was observed (Tables 2 to 5). Further, in the genotype B-3 group according to the present disclosure, the homology within the same group exhibited a relatively low homology at a level of about 95% to 100%. Thus, the possibility of subdividing the B-3 group into different genotype groups was identified.

(60) TABLE-US-00002 TABLE 2 Sequence homology (%) (Nucleotide identity) Genotype B-1 B-2 B-3 B-1 96.4~100.0% 96.0~97.4% 95.9~97.1% B-2 98.6~99.6% 97.2~100.0% 96.2~97.1% B-3 98.1~99.8% 98.2~99.9% 96.5~100.0%

Comparison of Homology of L Gene

(61) TABLE-US-00003 TABLE 3 Sequence homology (%) (Nucleotide identity) Genotype B-1 B-2 B-3 B-1 95.8~100.0% 94.5~97.1% 93.2~96.8% B-2 98.3~99.6% 95.7~100.0% 93.7~96.8% B-3 94.8~99.3% 95.3~99.7% 94.4~100.0%

Comparison of Homology of M Gene

(62) TABLE-US-00004 TABLE 4 Sequence homology (%) (Nucleotide identity) Genotype B-1 B-2 B-3 B-1 95.8~100.0% 94.5~97.1% 93.2~96.8% B-2 98.3~99.6% 95.7~100.0% 93.7~96.8% B-3 94.8~99.3% 95.3~99.7% 94.4~100.0%

Comparison of Homology of S (NP) Gene

(63) TABLE-US-00005 TABLE 5 Sequence homology (%) (Nucleotide identity) Genotype B-1 B-2 B-3 B-1 95.8~100.0% 94.8~97.4% 94.3~96.8% B-2 98.9~100.0% 95.1~100.0% 94.8~98.2% B-3 96.2~100.0% 94.9~100.0% 96.1~100.0%

Comparison of Homology of S (NS) Gene

Example 5

Cross Immunogenic Analysis and Vaccine Effect Identification Based on Genotype

(64) 5-1. Cross Immunogenic Analysis Based on Genotype

(65) In order to compare and analyze cross immunogenicity based on each genotype, a fifty percent of focus reduction neutralization test (FRNT50) was performed. Specifically, we carried out mass

proliferation of viruses of the new subdivided genotype B-1, B-2 and B-3 as isolated according to the present disclosure, and then carried out inactivation thereof by adding formalin (0.05%) thereto, and then identified occurrence or non-occurrence of the inactivation thereof via 3 times virus isolations. The inactivated whole vaccines were used to produce proteins via ultracentrifugation using 20% sucrose. The proteins were immunized into ferrets. After 2 weeks, additional immunization was performed on the ferrets (two times immunizations, 2 weeks intervals), and the blood was collected therefrom and the serum was separated therefrom. The separated serum was inactivated at 56° C. for 30 minutes, diluted to 1/10, and then serially diluted 2 times. The virus diluted with 200 FFU/ml was reacted with the virus as serially diluted at 37° C. at 1:1. After washing the VeroE6 cells as dispensed in a 6-well plate, the cells were infected with the reacted virus. One hour thereafter, we performed washing of the cells. Then, the cells were immersed in 0.8% DMEM agarose gel containing 1% FBS. 5 days after the infection, formalin was used to fix the cells, and 3 hours thereafter, three washes were performed, followed by treatment with 10% triton x-100 for 5 minutes at room temperature. Then, the cells were washed 3 times and blocking thereof was performed with 5% BSA. After incubation of the cells using a produced polyclonal NP antibody as a primary antibody, the cells were washed three times, and the cells were reacted with the HRP conjugated antibody as a secondary antibody for 1 hour, and then were washed, and then colored with DAB to identify the coloring result. The result was interpreted such that up to a value reduced by 50% compared to that of the focus forming of the well infected with only the virus was effective.

(66) TABLE-US-00006 TABLE 6 Serum Virus B-1 B-2 B-3 B-1 1280 1280 320 B-2 640 2560 320 B-3 640 640 640

(67) It was found based on the result of identifying cross-immunogenicity of the B-1, B-2, and B-3, the highest titer was found for each virus as shown in the Table 6. Thus, the viruses belonging to the same genotype exhibited high cross-neutralization reactivity, but the viruses belonging to different genotypes exhibited relatively low cross-neutralization reactivity.

(68) 5-2. Vaccine Effectiveness Identification

(69) After proliferating the viruses of the new subdivided genotypes B-1, B-2 and B-3 isolated according to the present disclosure in large quantities, formalin (0.05%) was added thereto to inactivate each virus. Then, whether each virus was successfully inactivated was checked via three times virus isolations. Each of the inactivated whole vaccine was immunized into 5 ferrets twice at 2 weeks intervals, and the ferrets were challenged at $1 \times 10^{7.6}$ /ml for each virus.

(70) It was identified based on the result of the attack challenge that the control animals died within 10 days of all challenge attack virus infections, but all ferrets of the vaccinated group survived.

Body temperature increase and weight loss were identified 2 to 8 days after the infection.

Thereafter, they were recovered (FIG. 6).

(71) Summarizing the above results, there are various genotypes of viruses having various genes in SFTSV. The viruses belonging to the same genotype exhibited the relatively high gene homology and high cross-immune response, but the viruses of the different genotypes exhibited the relatively low gene homology and low cross-immunogenicity. Thus, it may be inferred that in order to exhibit the cross-immunogenicity between the various genotypes, only a specific genotype of virus may exhibit limited protective ability.

(72) Therefore, the new viruses CB3/2016, CB7/2017 and CB6/2016 belonging to the subdivided genotypes B-1, B-2 and B-3 according to the present disclosure respectively are useful as vaccines having excellent cross immunogenicity with the SFTSV of the genotype B.

Claims

1. An immunogenic composition for prevention or treatment of a severe fever with thrombocytopenia syndrome, comprising inactivated severe fever with thrombocytopenia

syndrome virus or a gene thereof as an active ingredient, and an adjuvant, wherein the severe fever with thrombocytopenia syndrome virus or the gene thereof comprises: (i) a L gene that encodes a protein having the sequence of SEQ ID NO: 13; (ii) a M gene that encodes a protein having the sequence of SEQ ID NO: 14, (iii) a NP gene that encodes a protein having the sequence of SEQ ID NO: 15 and (iv) a NS gene that encode a protein having the sequence of SEQ ID NO: 16.

2. The composition of claim 1, wherein the severe fever with thrombocytopenia syndrome virus contains an L gene containing a base sequence represented by SEQ ID NO: 1, an M gene containing a base sequence represented by SEQ ID NO: 2, an NP gene containing a base sequence represented by SEQ ID NO: 3, and an NS gene containing a base sequence represented by SEQ ID NO: 4.

3. The composition of claim 1, wherein the composition further comprises a pharmaceutically acceptable carrier.

4. A method for detecting a severe fever with thrombocytopenia syndrome virus antibody, the method comprising: (a) contacting a sample isolated from a specimen with severe fever with thrombocytopenia syndrome virus or a gene thereof under a condition in which an antigen-antibody complex is able to be formed; and (b) detecting the formation of the antigen-antibody complex, wherein the severe fever with thrombocytopenia syndrome virus, or the gene thereof comprises (i) a L gene that encodes a protein having the sequence of SEQ ID NO: 13; (ii) a M gene that encodes a protein having the sequence of SEQ ID NO: 14, (iii) a NP gene that encodes a protein having the sequence of SEQ ID NO: 15 and (iv) a NS gene that encode a protein having the sequence of SEQ ID NO: 16.

5. A method for producing antiserum against a severe fever with thrombocytopenia syndrome virus in a non-human animal, the method comprising: (a) administering severe fever with thrombocytopenia syndrome virus or a gene thereof to the non-human animal at an amount effective to induce an immune response; and (b) collecting antiserum or plasma containing an antibody against the severe fever with thrombocytopenia syndrome virus, wherein the severe fever with thrombocytopenia syndrome virus or the gene thereof comprises (v) a L gene that encodes a protein having the sequence of SEQ ID NO: 13; (vi) a M gene that encodes a protein having the sequence of SEQ ID NO: 14, (vii) a NP gene that encodes a protein having the sequence of SEQ ID NO: 15 and (viii) a NS gene that encode a protein having the sequence of SEQ ID NO: 16.
